Hyperspectral Images Single-Source Domain Generalization Based on Nonlinear Sample Generation

Biqi Wang, *Graduate Student Member, IEEE*, Yang Xu, *Member, IEEE*, Zebin Wu, *Senior Member, IEEE*, Shangdong Zheng, Zhihui Wei, *Member, IEEE*, and Jocelyn Chanussot, *Fellow, IEEE*

***Abstract*— In hyperspectral cross-scene classification tasks, it is often challenging to obtain target domain samples during the training phase. Therefore, models need to be trained on one or multiple source domains and achieve good generalization performance on unknown target domains, known as domain generalization. The presence of domain shift limits the model's generalization across different domains, while the unknown target domain makes it difficult to accurately characterize the distribution differences between domains. To address this issue, we propose a generalization network based on nonlinear sample generation. The network divides the sample features into invariant features and variant features and generates samples by applying nonlinear transformations to the variant features. To ensure the quality of the generated samples, we introduce contrastive learning into the model. It ensures consistency in similarity between the generated samples and the source samples while maintaining a certain degree of dissimilarity. Experiments conducted on four cross-domain adaptive scenarios demonstrate the superior performance of our proposed method.**

***Index Terms*— Domain generalization (DG), hyperspectral image (HSI) classification, spectral unmixing.**

I. Introduction

With the advent of deep learning, convolutional neural networks (CNNs) have gained considerable attention and demonstrated remarkable performance in hyperspectral image (HSI) classification. Deep learning methods heavily rely on large amounts of labeled data for training, which is a laborious and expensive task for annotating HSIs [1]. As a result, it is necessary to effectively reuse existing labeled data. This means that the model trained on the existing annotated data should perform well on other unlabeled datasets. Hang et al. [2] use multimodal data for unsupervised feature learning and adopt a dual fine-tuning strategy to achieve knowledge transfer.

Due to various factors such as environmental conditions, acquisition devices, and lighting conditions [3], spectral reflectance differences can arise among different hyperspectral datasets. Traditional hyperspectral classification methods, which assume that the training and testing sets are from the same distribution, do not yield satisfactory results when dealing with cross-scene classification problems. To reduce the impact of noise, [4] introduced a knowledge distillation framework for weather-independent building segmentation. Su et al. [5] integrate a meta-heuristic optimization strategy to refine the graph structure of graph neural networks (GNNs). Hang et al. [6] address the issue of objects with varying scales by cascading three subnetworks.

To tackle the challenge of distribution discrepancy in HSI classification, domain adaptation methods have been introduced. We refer to the labeled training set as the source domain, and the unlabeled testing set as the target domain. The existing domain adaptation methods can generally be categorized into two types [7]: one focuses on selecting appropriate samples from both the source and target domains to minimize the distance between them, while the other aims to find a feature subspace that reduces the distribution differences between the two domains. Both types of methods require the target domain data to be visible during the training process. However, in certain scenarios, it is not feasible to access the target domain data. For example, all the samples of a specific land cover class under different weather conditions or all the medical images of patients with different symptoms. In such cases, the concept of domain generalization (DG) has been proposed.

The goal of DG is to train a model using one or multiple diverse but related source domain datasets, so that the model can perform well on unknown target domains. Depending on the number of source domains, DG can be categorized as single-source DG and multisource DG. The existing DG methods can be broadly classified into three categories: data

Manuscript received 6 February 2024; revised 9 April 2024; accepted 19 April 2024. Date of publication 29 April 2024; date of current version 7 May 2024. This work was supported in part by the National Natural Science Foundation of China under Grant U23B2006, Grant 92370203, and Grant 62071233; in part by Jiangsu Provincial Innovation Support Program under Grant BZ2023046; in part by Jiangsu Provincial Natural Science Foundation of China under Grant BK20211570 and Grant BK20170858; in part by Jiangsu Provincial Key Research and Development Program under Grant BE2022065-2; and in part by the Open Foundations of Jiangsu Province Engineering Research Center of Airborne Detecting and Intelligent Perceptive Technology under Grant JSECF2023-03. *(Corresponding author: Zebin Wu.)*

Biqi Wang, Yang Xu, Zebin Wu, and Zhihui Wei are with the School of Computer Science and Engineering, Nanjing University of Science and Technology (NJUST), Nanjing 210094, China (e-mail: wangbiqi@njust.edu.cn; xuyangth90@njust.edu.cn; zebin.wu@gmail.com; gswei@njust.edu.cn).

Shangdong Zheng is with the School of Computer Science, Nanjing University of Posts and Telecommunications, Nanjing 210023, China (e-mail: zhengsd@njupt.edu.cn).

Jocelyn Chanussot is with Inria, CNRS, Grenoble INP, LJK, Université Grenoble Alpes, 38000 Grenoble, France (e-mail: jocelyn.chanussot@grenoble-inp.fr).

Digital Object Identifier 10.1109/TGRS.2024.3394435

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:59:23 UTC from IEEE Xplore. Restrictions apply.

processing, representation learning, and learning strategies. Data processing involves transforming or enhancing the source domain data to improve the model's generalization performance, primarily applied in single-source DG. Representation learning focuses on learning domain-invariant features that enable the model to adapt well to different scenarios, primarily used in multisource DG. Learning strategies involve incorporating mature learning patterns from machine learning into multidomain training to enhance the model's generalization ability.

DG methods have been widely applied in various fields, such as medical analysis, natural language processing, fault detection, and more. Recently, there has been a focus on DG problems in the context of HSI classification. Compared with typical domain adaptation methods, SDEnet [8] has been shown to be more practical for DG. It uses a semantic encoder and a morphological encoder to generate samples with distribution differences from the source domain while ensuring the validity of the generated samples. LDGnet [9] incorporates text information as prior knowledge shared across domains. They design image encoders and text encoders to learn domain-invariant representations. LLURnet [10] proposes a dual adversarial regularization training approach for both the generator and the discriminator. This approach aims to extract spectral information from images and avoids excessive overlap between generated samples and source domain samples.

To address the issue of poor generalization performance of the model on unknown target domains, we need to generate samples that are similar but not identical to the source domain data. The source and target domains share domain-invariant features, while the domain-variable features contain domain-specific information. Therefore, we can decompose the source domain samples into domain-invariant features and domain-variable features and generate effective samples by transforming the domain-variable features. Spectral unmixing assumes that there are differences in endmember information between samples from different domains, while samples of the same class have similar abundance matrices. In real-world scenarios, mixed pixels are influenced by factors such as occlusion and nonlinear reflection. Nonlinear unmixing can more accurately capture these nonlinear characteristics, model the mixing process, and provide more accurate unmixing results. Based on a general mixing mechanism [11], we design an encoder–decoder architecture to extract invariant and variant features. We introduce random environmental noise that changes with the environment. By adding noise to the linear part of the variant features, the extension domain can be generated. However, due to the randomness of the generation process, the generated domain samples may resemble the source domain samples, causing information redundancy. Moreover, they may not correspond to the labels, leading to classification errors. The two scenarios can hinder the improvement of generalization performance and even have negative effects. To ensure the effectiveness of the generated samples, we introduce contrastive learning, which ensures differences between the generated samples and the source domain samples, while maintaining similarity in invariant features among different-domain samples of the same class. In this article, our contributions can be divided into the following three aspects.

1) Based on the idea of nonlinear spectral unmixing model, we designed an encoder–decoder architecture to separate invariant features and variant features.
2) We designed a random environmental noise to simulate the differences between different domains and generated new samples by adding noise to the linear components.
3) By introducing contrastive learning, we ensure the effectiveness of generated samples by considering both their differences and connections with source domain samples.

In the rest of this article, we first present an overview of the related work in Section II. We provide a detailed explanation of our method in Section III. Extensive experiments conducted on four scenarios show the performance of our method in Section IV. Section V provides a summary of our work.

II. Related Works

A. Domain Generation

Traditional domain adaptation methods require visible target domain samples during the training phase, but in some scenarios, it is not possible to acquire target domain data in advance. Therefore, the concept of DG is proposed, aiming to learn a model from one or multiple different but related datasets and exhibit good generalization performance on unknown target domains. The existing DG methods can be roughly categorized into three types [7]: data manipulation, representation learning, and learning strategies.

The generalization performance of machine learning relies on sufficient training data, and in the DG problem with only limited source domain samples, data manipulation is one of the lowest cost means for sample generation. Some researchers try to other types of data. Mei et al. [12] introduce a unified framework that jointly enhances spatial resolution and spectral resolution to make full use of multispectral data. To reconstruct HSI using RGB inputs, [13] designs a two-stage neural network based on the unique characteristics of HSI, while Mei et al. [14] propose a lightweight feature extraction module. Hang et al. [15] design a spectral super-resolution network to use two intrinsic properties, such as spectral correlation and projection property. The primary objective of data manipulation-based DG methods is to use techniques such as data augmentation or data generation to enhance the diversity of THE existing training data. Peng et al. [16] propose an adversarial method based on meta-learning to achieve domain expansion and integrated uncertainty quantification to ensure effectiveness. DDAIG [17] uses adversarial learning to train a sample generation network, enhancing the robustness of the classifier to unknown target domains. Somavarapu et al. [18] narrow the domain gap through image-style transformation. Mixup [19] generates new samples by linearly interpolating between any two source domain samples, where the weights for interpolation are drawn from a Beta distribution.

Representation learning improves the model's generalization performance by learning shared feature representations that can adapt to different domains. to achieve a more reliable

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:59:23 UTC from IEEE Xplore. Restrictions apply.

feature representation, [20] integrates nonlocal spatial similarity and local spectral similarity. Su et al. [21] capture local-to-global correlations through normalized spectral clustering and kernel learning. Some researchers try to learn domain-invariant representations. Considering that there may be differences in both the marginal and conditional distributions between different domains, Li et al. [22] propose learning feature representations that are invariant to the domain's conditional distribution. To minimize the feature differences between domains, Rahman et al. [23] design a framework that uses correlation alignment and adversarial learning. Invariant risk minimization (IRM [24]) estimates nonlinear and invariant causal factors from multiple training datasets. Some researchers have also attempted to decompose sample features. Common specific decomposition (CSD [25]) jointly learns a common component and a domain-specific component through low-rank decomposition. Zhang et al. [26] simultaneously learn semantic and variational autoencoders to achieve feature disentanglement. Liu et al. [27] use causal invariance to achieve DG and theoretically prove the feasibility of this method.

In addition, DG has also been studied in the general machine learning paradigm. Zhou et al. [28] design a model with multiple classifier heads. This model improves generalization performance by leveraging complementary information from multiple source domains through ensemble learning. Kim et al. [29] propose a self-supervised contrastive regularization method that effectively applies mixed augmentation using only positive pairs of data. Narayanan et al. [30] represent the shape content of images using the shock graph of image contours and use graph convolutional networks for classification.

B. Spectral Unmixing

Due to the low spatial resolution of hyperspectral imaging devices, the diversity of materials in the captured scenes, and the spectral reflections from different objects, the spectrum of a single pixel in an HSI is typically a mixture of multiple materials. To better analyze remote sensing data, it is necessary to decompose individual pixels into combinations of endmembers and determine the mixing coefficients, which are represented by the abundance matrix. Among the available spectral unmixing models, the most commonly used one is the linear mixing model (LMM [31])

$$\boldsymbol{x} = \boldsymbol{M}\boldsymbol{a} + \boldsymbol{n} \tag{1}$$

where $\boldsymbol{x} \in R^B$, B denotes the number of spectral bands. $\boldsymbol{M} = [m_1, m_2, \ldots, m_R]$ is the endmember matrix, and R represents the number of endmembers. $\boldsymbol{a} = [a_1, a_2, \ldots, a_R]^T$ represents the abundance vector, and $\boldsymbol{n} \in R^B$ denotes the adaptive noise vector.

Due to physical constraints, abundances need to satisfy nonnegativity constraint [32] and sum-to-one constraint [33]

$$\forall i : \boldsymbol{a}_i \geqslant 0 \tag{2}$$

$$\sum_{i=1}^{R} \boldsymbol{a}_i = 1. \tag{3}$$

LMM assumes that incident light is reflected by each component in the scene only once before being captured by the imaging device. Therefore, the observed spectrum can be represented as a linear combination of each component. In the early stages, blind source separation methods were primarily used, with independent component analysis (ICA [34]) being the most classic approach, which separates independent components for unmixing. Based on the shared similar smooth structures between similar patches in HSIs, Yao et al. [35] introduce total variation regularization to capture spatial information in the image. Hong et al. [36] model spectral variability using an endmember dictionary and learned a spectral variation dictionary while performing abundance estimation.

In real-world scenarios, incident light may undergo complex interactions between multiple materials, and a simple LMM may not suffice. Therefore, many studies have been dedicated to addressing the problem of nonlinear unmixing. Borel and Gerstl [37] propose a bilinear model that incorporates bilinear interaction terms into the linear synthetic spectrum. In recent years, with the development of deep learning, neural-network-based spectral unmixing methods have gained widespread attention. Licciardi and Del Frate [38] use a neural network model to map the input vector to the abundance matrix. Palsson et al. [39] propose a high-dimensional blind spectral unmixing method based on autoencoders. Qi et al. [40] improve model performance by combining spatial and spectral information. Palsson et al. [41] apply CNNs to spectral unmixing, using the spectral structure of HSIs for endmember and abundance estimation. Zhao et al. [11] propose a general mixing mechanism as follows:

$$\boldsymbol{x} = \boldsymbol{M}\boldsymbol{a} + \Psi(\boldsymbol{M}, \boldsymbol{a}) + \boldsymbol{n} \tag{4}$$

where $\Psi(\cdot)$ represents a nonlinear mixing function between endmembers and abundance coefficients.

It used an autoencoder to learn this non LMM. Therefore, the model can be equivalently transformed as

$$\boldsymbol{x} = \Phi(\boldsymbol{M}_D\boldsymbol{a}) + \Psi(\boldsymbol{M}_D\boldsymbol{a}) + \boldsymbol{n} \tag{5}$$

where $\Phi(\cdot)$ represents stepwise summation operator, and $\boldsymbol{M}_D$ is a diagonal matrix with its diagonal elements representing R endmember vectors corresponding to B spectral bands.

III. Methods

A. Problem Formulation

For conventional domain adaptation, the source domain is defined as $D_S = \{\boldsymbol{x}_i^S, \boldsymbol{y}_i^S\}_{i=1}^{N_S}$, and $\boldsymbol{x}_i^S$ and $\boldsymbol{y}_i^S$ represent the source images and labels, respectively. The target domain is defined as $D_T = \{\boldsymbol{x}_i^T\}_{i=1}^{N_T}$. There is a distribution discrepancy between the source domain and the target domain data, $P(X_S) \neq P(X_T)$. In the DG problem, the target domain data are not available during the training stage, meaning that only D_S is provided for training.

Our proposed network consists of a sample generation module and a domain alignment module, as shown in Fig. 1. We select fixed-sized patches from the HSI. The patches are input into the encoder to compress into a low-dimensional representation as domain-invariant features. The decoder

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:59:23 UTC from IEEE Xplore. Restrictions apply.

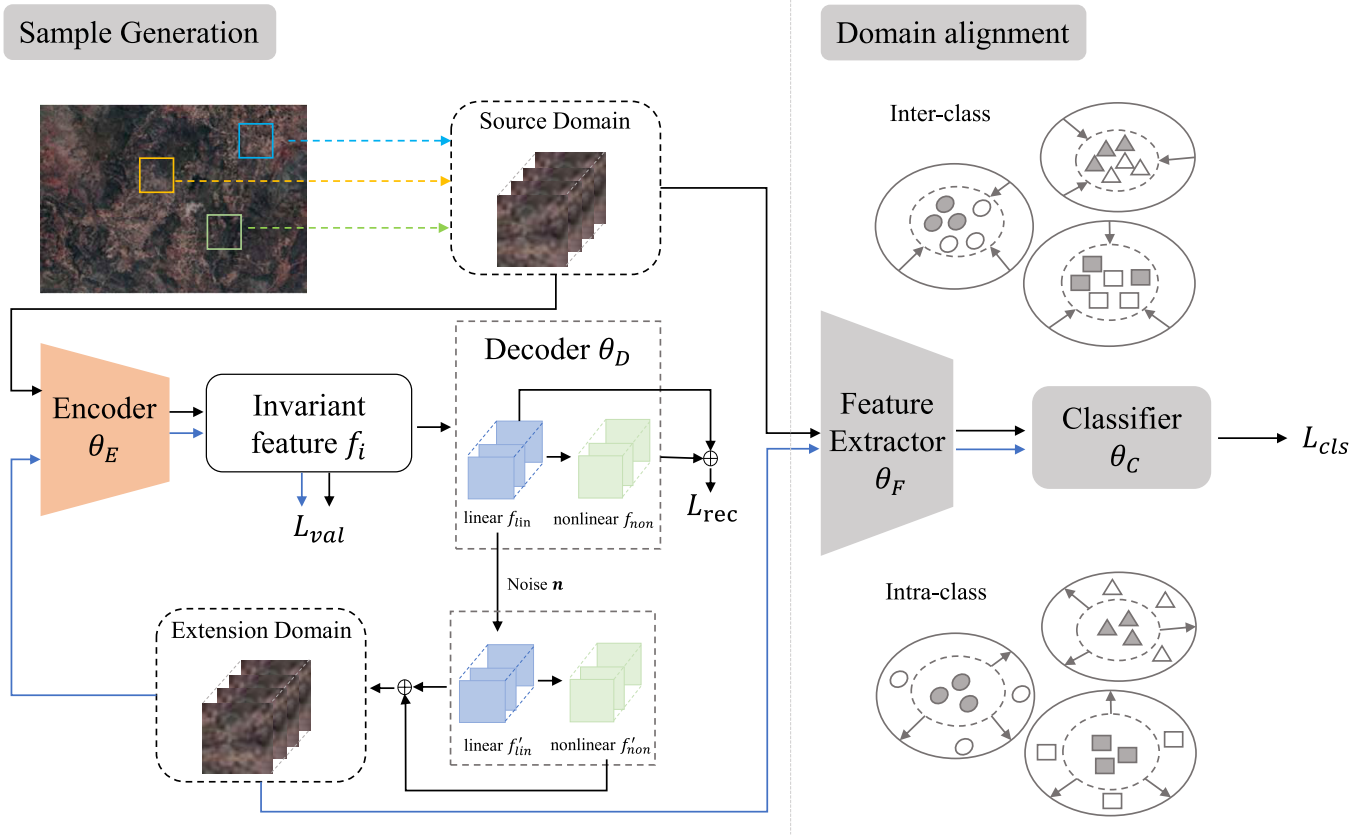


Fig. 1. Framework of the proposed model. The model consists of a sample generation module and a domain alignment module. The sample generation consists of an encoder and a decoder. The encoder is used to extract the invariant features. The decoder is used to reconstruct samples. The extended domain is generated by introducing noise to the linear components. The circles, triangles, and squares represent different classes.

reconstructs the original input by decompressing the low-dimensional representation, and the model parameters are trained by minimizing the error between the reconstructed result and the original input. The decoder features are divided into linear and nonlinear components. The extension domain samples are generated by adding noise to the linear part. The source domain and extension domain samples are then input to the feature extraction network for domain alignment. The output of the feature extraction network is used as input to the classifier, and the cross-entropy loss is computed to facilitate sample classification.

B. Invariant and Variable Feature Decomposition

In the single-source DG problem, there are distribution differences between the training set and the test set due to factors such as data collection devices, environments, and time. Therefore, models trained only on the source domain data tend to have poor generalization performance on unknown target domains. To address this issue, it is crucial to extract domain-invariant features from the single-source domain data and enhance the model's adaptability across different domains.

For HSIs, assuming that there are differences between endmembers in different domains due to domain shift, pixels belonging to the same class but in different domains have a similar abundance matrix [42]. Based on this assumption, we can treat the extraction of invariant features as a spectral unmixing problem.

The commonly used linear mixture model assumes that the incident light is reflected only once by the materials in the scene before image acquisition. However, in the real world, the incident light undergoes multiple reflections between different materials in complex scenes. A simple linear model cannot accurately describe the relationship between endmembers, mixing coefficients, and hyperspectral pixels. To achieve spectral unmixing in complex environments, we use a general spectral mixture model that consists of both linear mixture components and additional nonlinear mixture components [11].

We consider the abundance matrix $\boldsymbol{M}$ can be seen as the invariant feature $\boldsymbol{f}_i$, and the endmember information $\boldsymbol{a}$ can be seen as the variant feature $\boldsymbol{f}_v$

$$\begin{cases} \boldsymbol{M} \Rightarrow \boldsymbol{f}_i \\ \boldsymbol{a} \Rightarrow \boldsymbol{f}_v. \end{cases} \tag{6}$$

According to the nonlinear spectral unmixing model (5), we can derive the following equation:

$$\boldsymbol{x} = \Phi\left(\boldsymbol{f}_v \boldsymbol{f}_i\right) + \Psi\left(\boldsymbol{f}_v \boldsymbol{f}_i\right) + \boldsymbol{n} \tag{7}$$

where $\boldsymbol{f}_i$ needs to satisfy nonnegativity constraint and sum-to-one constraint.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:59:23 UTC from IEEE Xplore. Restrictions apply.

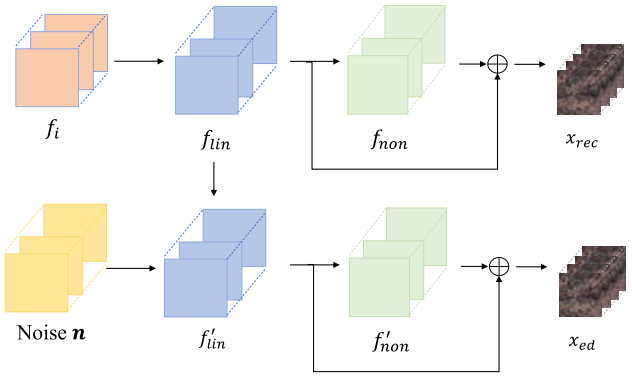


Fig. 2. Process of the sample generation. The transformed linear and nonlinear components are used together to generate target domain samples.

The encoder E compresses the input samples $\boldsymbol{x}_s$ to a low-dimensional representation $\boldsymbol{z}$

$$\boldsymbol{z} = \boldsymbol{\theta}_E(x_s). \tag{8}$$

The decoder D reconstructs the low-dimensional representation $\boldsymbol{z}$ to obtain a reconstructed output $\hat{\boldsymbol{x}}_s$, that is,

$$\hat{\boldsymbol{x}}_s = \boldsymbol{\theta}_D(\boldsymbol{z}). \tag{9}$$

By minimizing the mean squared error (mse) between the reconstructed images $\boldsymbol{x}_{\text{inter}} = \boldsymbol{\theta}_D(\boldsymbol{\theta}_E(\boldsymbol{x}_s))$ and the input samples $\boldsymbol{x}_s$, the parameters of the autoencoder can be trained. The reconstruction loss can be achieved as follows:

$$L_{\text{inter}} = \frac{1}{N}\sum_{i=1}^{N} \|\boldsymbol{x}_{\text{inter}} - \boldsymbol{x}_s\|^2 \tag{10}$$

where N means the number of the input pixels.

If we denote the network parameters of the first layer of the decoder as $\boldsymbol{\theta}_1$, then the reconstruction process of the decoder can be written in the following format:

$$\boldsymbol{x} = \Phi(\boldsymbol{\theta}_1 z) + \Psi(\boldsymbol{\theta}_1 z) = \boldsymbol{f}_{\text{non}} + \boldsymbol{f}_{\text{non}}. \tag{11}$$

According to (7) and (11), we can treat the output of encoder E as $\boldsymbol{f}_i$ and the first layer parameters $\boldsymbol{\theta}_1$ of the decoder as $\boldsymbol{f}_v$

$$\begin{cases} z \Rightarrow \boldsymbol{f}_i \\ \boldsymbol{\theta}_1 \Rightarrow \boldsymbol{f}_v. \end{cases} \tag{12}$$

We use the decoder to simulate the output of the general mixture model. To satisfy the nonnegativity constraint, we enforce absolute value calculation on the results extracted by the encoder. In addition, to ensure that their sum is equal to 1, the absolute values are normalized

$$\hat{\boldsymbol{f}}_i = \frac{|\boldsymbol{f}_i|}{\sum_{k=1}^{R} |\boldsymbol{f}_i^k|}. \tag{13}$$

C. Sample Generation

Traditional domain adaptation methods often rely on either selecting similar samples from the source and target domains to minimize the distribution discrepancy or mapping samples to a common feature space for distribution alignment. Both the methods require access to target domain samples during the training stage. However, in the case of single-source DG problem, where target domain samples are unavailable beforehand, the distribution discrepancy between training samples and target task samples cannot be accurately described. As a result, traditional domain adaptation methods are not applicable to this problem.

To improve the generalization performance of the model on unknown target domains, we generate new samples with distribution discrepancies from the existing training data. According to (6), the features of source domain samples are decomposed into variant features $\boldsymbol{f}_v$ and invariant features $\boldsymbol{f}_i$. The invariant features represent the parts of the source domain samples that are insensitive to domain shift, while the variant features reflect the parts that are heavily influenced by domain shift. According to Fig. 2, we transform the variable feature distribution by adding noise to the linear components of the decoder. To facilitate the transformation, we introduce a random environment noise to simulate the difference between different domains

$$\boldsymbol{n}_r = \mathbb{N}(\mu_r, \sigma_r) \tag{14}$$

where $\boldsymbol{n}_r$ is a random noise that matches the shape of $\boldsymbol{f}_{\text{non}}$ and follows a normal distribution.

To avoid significant differences between the added noise and the original sample distribution, we assume that the original sample also follows a normal distribution. Then the distribution of the environmental noise $\boldsymbol{n}_e$ can be obtained by the linear combination of the original sample and the random noise distribution

$$\begin{cases} \mu_e = \alpha * \mu_r + (1-\alpha) * \mu_{f_{\text{non}}} \\ \sigma_e = \alpha * \sigma_r + (1-\alpha) * \sigma_{f_{\text{non}}} \end{cases} \tag{15}$$

where α is a trainable parameter.

The transformed linear components $\boldsymbol{f}'_{\text{non}}$ can be obtained by the linear combination of $\boldsymbol{f}_{\text{non}}$ and $\boldsymbol{n}_e$

$$\boldsymbol{f}'_{\text{non}} = \beta \boldsymbol{n}_e + (1-\beta)\boldsymbol{f}_{\text{non}} \tag{16}$$

where β is a trainable parameter.

The corresponding nonlinear component $\boldsymbol{f}'_{\text{non}}$ can be obtained by $\boldsymbol{f}'_{\text{non}}$. Then, we can generate an extension domain based on (11)

$$\boldsymbol{x}_e = \boldsymbol{f}'_{\text{non}} + \boldsymbol{f}'_{\text{non}}. \tag{17}$$

D. Validity Constraints

Due to the influence of randomization, the quality of generated samples can vary, and it inevitably leads to the presence of some invalid samples. Invalid samples can be classified into two categories. The first category includes samples that are too similar to the source domain samples. As the distribution difference between the source domain and the expanded domain is small, these samples may not effectively enhance the model's generalization performance and even increase the risk of overfitting. The second category includes samples that are significantly different from the source domain samples. During

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:59:23 UTC from IEEE Xplore. Restrictions apply.

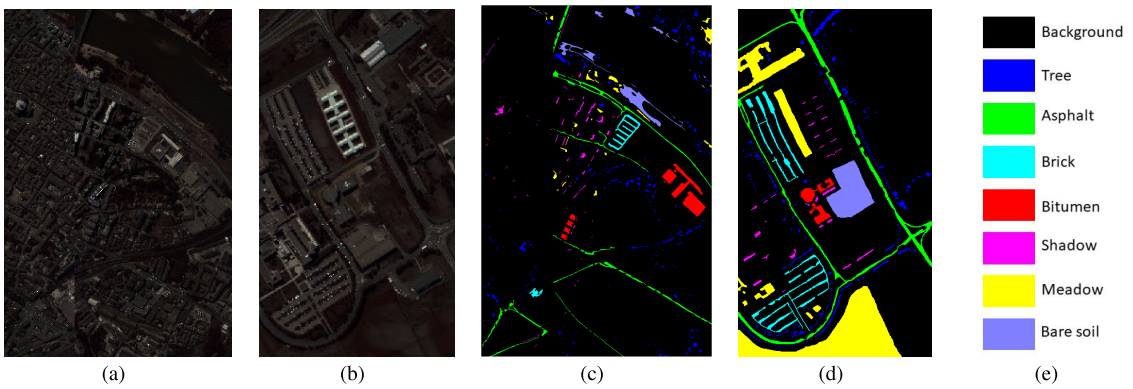


Fig. 3. Pavia. (a) PC false-color image. (b) UP false-color image. (c) PC ground-truth map. (d) UP ground-truth map. (e) sample categories.

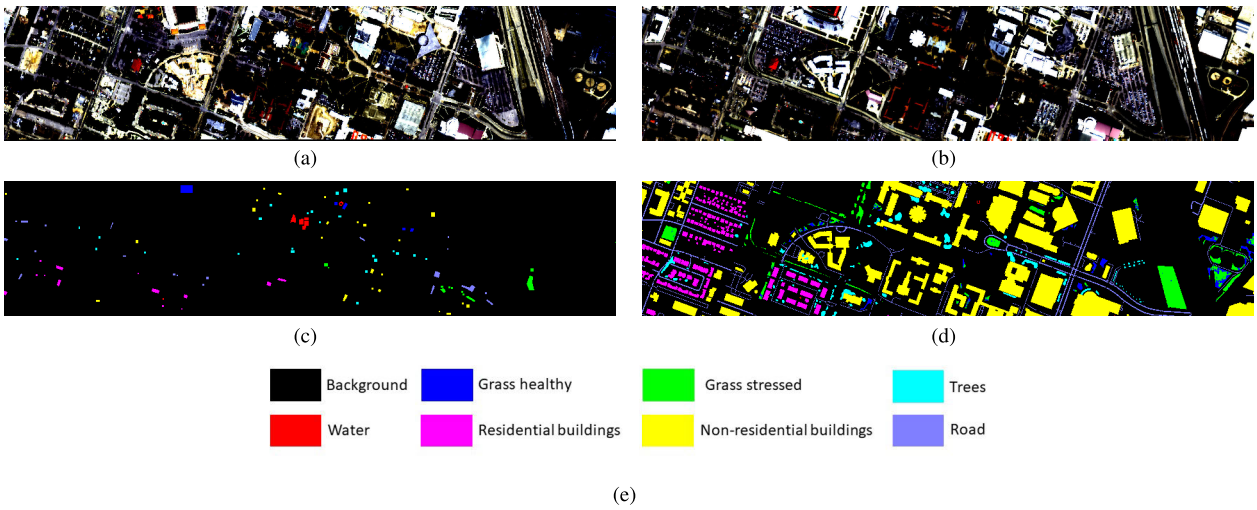


Fig. 4. Houston. (a) Houston 2013 false-color image. (b) Houston 2018 false-color image. (c) Houston 2013 ground-truth map. (d) Houston 2018 ground-truth map. (e) sample categories.

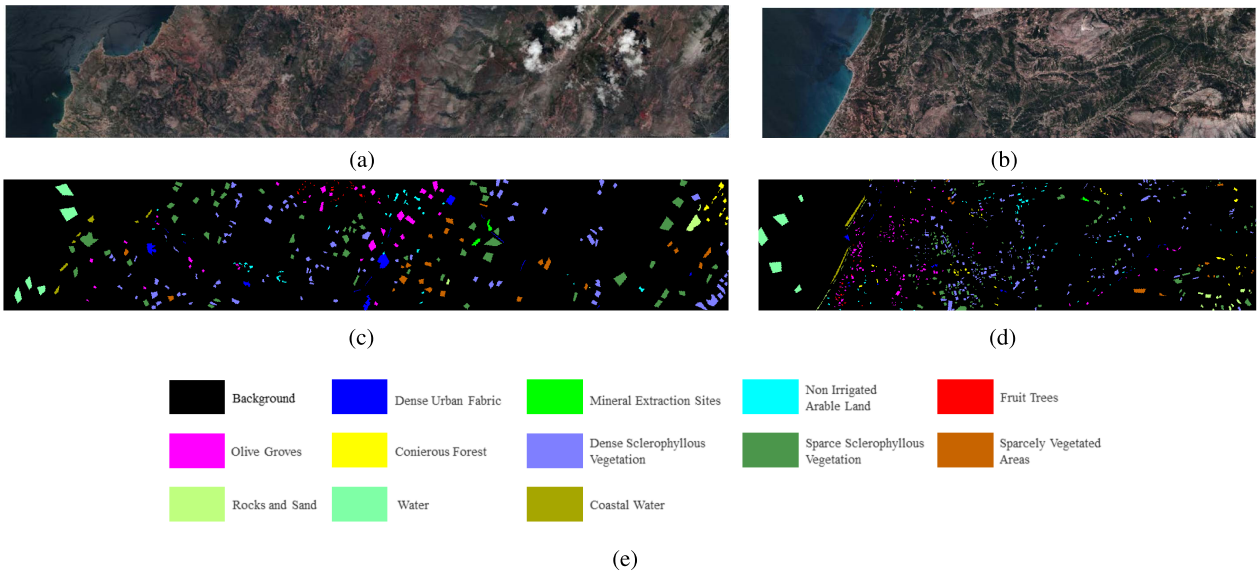


Fig. 5. HyRANK. (a) Dioni false-color image. (b) Loukia false-color image. (c) Dioni ground-truth map. (d) Loukia ground-truth map. (e) sample categories.

the generation process, these samples may undergo changes in their class information, which can mislead the model and result in a negative impact by incorrectly assigning these samples to the wrong classes.

We introduce contrastive learning to ensure the validity of generated samples. To avoid significant differences between the source domain and the extension domain, we hope that $\boldsymbol{f}_i$ extracted from samples from different domains of the same

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:59:23 UTC from IEEE Xplore. Restrictions apply.

TABLE I
SAMPLE DISTRIBUTION OF THE PAVIA DATASET

Class	Name	UP	PC
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadow	18649	3090
7	Bare soil	5029	6584

TABLE II
SAMPLE DISTRIBUTION OF THE HOUSTON DATASET

Class	Name	Houston 2013	Houston 2018
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365

TABLE III
SAMPLE DISTRIBUTION OF THE HYRANK DATASET

Class	Name	Dioni	Loukia
1	Dense Urban Fabric	1262	206
2	Mineral Extraction Sites	204	54
3	Non Irrigated Arable Land	614	426
4	Fruit Trees	150	79
5	Olive Groves	1768	1107
6	Conierous Forest	361	422
7	Dense Sclerophyllous Vegetation	5035	2996
8	Sparce Sclerophyllous Vegetation	6374	2361
9	Sparcely Vegetated Areas	1754	399
10	Rocks and Sand	492	453
11	Water	1612	1393
12	Coastal Water	398	421

class are as consistent as possible, thereby obtaining intraclass contrastive loss

$$L_{\text{intra}} = -\sum_{k\in D_s} \log\left(\frac{\sum_{m\in P(k)} d\left(\hat{\boldsymbol{f}}_{ik}, \hat{\boldsymbol{f}}_{im}\right)+\tau}{\sum_{n\in N(k)} d\left(\hat{\boldsymbol{f}}_{ik}, \hat{\boldsymbol{f}}_{in}\right)+\tau}\right) \tag{18}$$

where $P(k)$ represents a set that is consistent with the class of sample k, and $N(k)$ represents a set that is different with the class of sample k. τ is a minimum value that avoids a value of 0.

Meanwhile, the source domain and the generated extension domain should belong to different distributions, thus introducing interclass contrastive loss

$$L_{\text{intra}} = -\frac{1}{c}\sum_{c=0}^{C}\sum_{k\in D_s} \log\left(\frac{\sum_{m\in P_c(k)} d\left(\hat{\boldsymbol{f}}_{ik}, \hat{\boldsymbol{f}}_{im}\right)+\tau}{\sum_{n\in N_c(k)} d\left(\hat{\boldsymbol{f}}_{ik}, \hat{\boldsymbol{f}}_{in}\right)+\tau}\right) \tag{19}$$

where $P_c(k)$ represents the set of samples belonging to class c in the source domain. $N_c(k)$ represents the set of samples belonging to class c in the extension domain.

Therefore, the loss of sample validity is

$$L_{\text{val}} = L_{\text{intra}} + L_{\text{inter}}. \tag{20}$$

E. Domain Alignment Module

To achieve classification tasks, the source samples are input into the feature extractor $\boldsymbol{\theta}_F$ and classifier $\boldsymbol{\theta}_C$ to obtain prediction results. The classification loss is calculated using cross-entropy

$$L_{\text{cls}}^{\text{SD}} = -\sum_{i=1}^{N} \boldsymbol{y}_i \log(\boldsymbol{\theta}_F(\boldsymbol{\theta}_C(\boldsymbol{x}_s))). \tag{21}$$

In addition, we use a random parameter ε to linearly combine the source domain and the extended domain to obtain an intermediate domain (ID) to avoid significant differences between the two domains

$$\text{ID} = \varepsilon\text{SD} + (1-\varepsilon)\text{ED} \tag{22}$$

$$L_{\text{cls}}^{\text{ED}} = -\sum_{i=1}^{N} \boldsymbol{y}_i \log(\boldsymbol{\theta}_F(\boldsymbol{\theta}_C(\boldsymbol{x}_{ed}))) \tag{23}$$

$$L_{\text{cls}}^{\text{ID}} = -\sum_{i=1}^{N} \boldsymbol{y}_i \log(\boldsymbol{\theta}_F(\boldsymbol{\theta}_C(\boldsymbol{x}_{id}))). \tag{24}$$

By narrowing the differences between samples of the same class and expanding the differences between samples of different classes, distribution alignment is achieved, resulting in the class alignment loss

$$L_{\text{da}}^{\text{cls}} = -\sum_{k=0}^{n_c}\frac{1}{n_p}\sum_{m\in P(k)} \log\left(\frac{e^{d\left(\boldsymbol{\theta}_F(\boldsymbol{x}_k), \boldsymbol{\theta}_F\left(\boldsymbol{x}_m^+\right)\right)/\tau}}{\sum_{n\in N(k)} e^{d\left(\boldsymbol{\theta}_F(\boldsymbol{x}_k), \boldsymbol{\theta}_F\left(\boldsymbol{x}_n^-\right)\right)/\tau}}\right) \tag{25}$$

where n_p is the number of samples belonging to $P(k)$.

To optimize the sample generation module and domain alignment module, interdomain loss is introduced

$$L_{\text{da}}^{d} = -\sum_{c}\sum_{k=0}^{n_c}\frac{1}{n_{pk}}\sum_{m\in P_c(k)} \log\left(\frac{e^{d\left(\boldsymbol{\theta}_F(\boldsymbol{x}_k), \boldsymbol{\theta}_F\left(\boldsymbol{x}_m^+\right)\right)/\tau}}{\sum_{n\in N_c(k)} e^{d\left(\boldsymbol{\theta}_F(\boldsymbol{x}_k), \boldsymbol{\theta}_F\left(\boldsymbol{x}_n^-\right)\right)/\tau}}\right) \tag{26}$$

where n_{pk} is the number of samples belonging to $P_c(k)$.

In the training process, the sample generation module and domain alignment module are optimized separately. The objective of the domain alignment module is as follows:

$$\min_{\theta_F,\theta_C} L = L_{\text{cls}}^{\text{SD}} + L_{\text{cls}}^{\text{ED}} + L_{\text{cls}}^{\text{ID}} + L_{\text{da}}^{\text{cls}}. \tag{27}$$

The objective of the sample generation module is as follows:

$$\min_{\theta_E,\theta_D} L = L_{\text{cls}}^{\text{ED}} + L_{\text{val}} + \lambda_1 L_{\text{inter}} + \lambda_2 L_{\text{da}}^{d} \tag{28}$$

where λ_1 and λ_2 are hyperparameters.

IV. EXPERIMENTS

A. Empirical Setup

1) Datasets: To illustrate the effectiveness of our method, we conducted experiments in four cross-domain adaptive scenarios, Houston2013→Houston2018, UP→PC, PC→UP, and Dioni→Loukia.

The Pavia University (UP) is the hyperspectral data collected at the University of Pavia. The size of this dataset is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:59:23 UTC from IEEE Xplore. Restrictions apply.

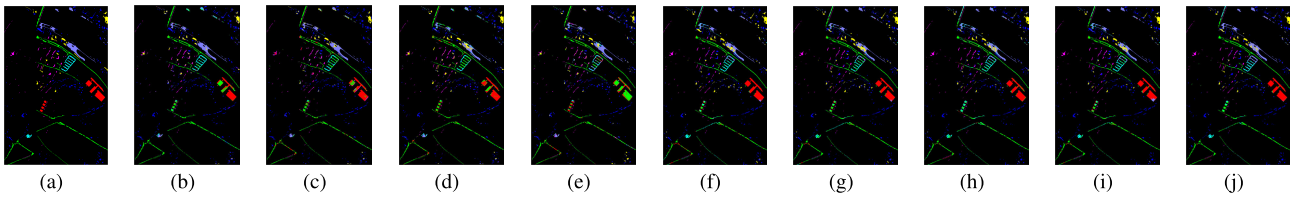


Fig. 6. PC. (a) Ground-truth map. (b)–(j) Classification maps for different methods. (b) Deep croal. (c) IRM. (d) VREX. (e) Mixstyle. (f) SagNET. (g) PDEN. (h) SDEnet. (i) LLURnet. (j) OURS.

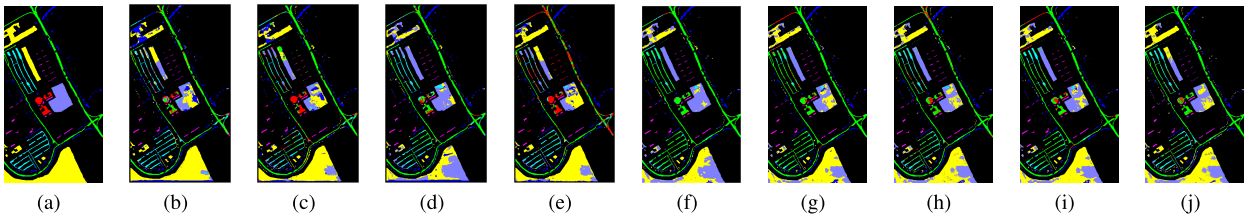


Fig. 7. UP. (a) Ground-truth map. (b)–(j) Classification maps for different methods. (b) Deep croal. (c) IRM. (d) VREX. (e) Mixstyle. (f) SagNET. (g) PDEN. (h) SDEnet. (i) LLURnet. (j) OURS.

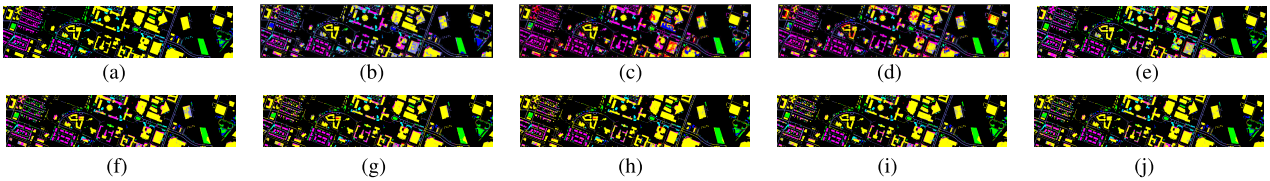


Fig. 8. Houston. (a) Ground-truth map. (b)–(j) Classification maps for different methods. (b) Deep croal. (c) IRM. (d) VREX. (e) Mixstyle. (f) SagNET. (g) PDEN. (h) SDEnet. (i) LLURnet. (j) OURS.

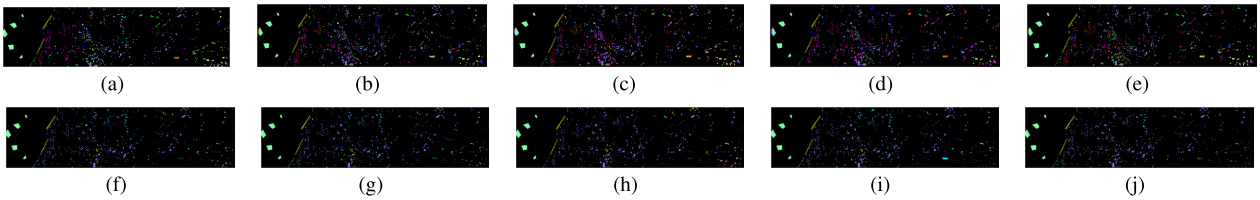


Fig. 9. Loukia. (a) Ground-truth map. (b)–(j) Classification maps for different methods. (b) Deep croal. (c) IRM. (d) VREX. (e) Mixstyle. (f) SagNET. (g) PDEN. (h) SDEnet. (i) LLURnet. (j) OURS.

610 × 340, and it contains 103 wavebands. The Pavia Center (PC) is the data collected in the center of Pavia City. Its size is 1096 × 715, including 102 wavebands. Two datasets are used as the source domain and the target domain, respectively. They contain seven shared classes. The information and the numbers of categories are shown in Table I. Fig. 3 shows their pseudo RGB images and ground-truth maps.

The Houston datasets were collected by different sensors on the campus of the University of Houston. Houston 2013 is 349 × 1905 in size, including 144 spectral bands, with a spatial resolution of 2.5 m. Houston 2018 is 209 × 955 in size, with only 49 spectral bands and a spatial resolution of 1 m. We take Houston 2013 as the source domain and Houston 2018 as the target domain. To make the dimensions of the source domain and the target domain consistent, we selected a 209 × 955 area in Houston 2013 and 48 spectral bands shared by the two datasets. The filtered dataset contains seven shared classes. The information and the numbers of categories are shown in Table II, and Fig. 4 shows their pseudo RGB images and ground-truth maps.

The Dioni is the hyperspectral data collected by Hyperion. The size of it is 250 × 1376, and it contains 176 wavebands. The size of Loukia is 249 × 945, including 176 wavebands. We take Dioni as the source domain and Loukia as the target domain. The two datasets contain 12 shared classes. The information and the numbers of categories are shown in Table III. Fig. 5 shows their pseudo RGB images and ground-truth maps.

2) Metrics: We use three indicators to evaluate the model: overall accuracy, average category accuracy, and kappa coefficient.

B. Implementation Details

We choose seven domain generation methods to validate the effectiveness of our method, IRM [24], VREx [43], Mixstyle [44], SagNet [45], PDEN [46], SDEnet [8], and LLURNET [10]. To compare the performance of domain adaptation methods and DG methods, we introduce a domain adaptation method Deep coral [47].

To match the input size of ResNet18, the patch size for SagNet is set to 32 × 32. For other methods, the patch size is uniformly set to 13 × 13. We divide the source domain dataset into a training set and a validation set, randomly selecting 80% of the samples as the training set and the remaining 20% as the testing set.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:59:23 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
CLASSIFICATION RESULTS (%) OF THE PC DATASET

Class	Deep coral	IRM	VREx	Mixstyle	SagNet	PDEN	SDEnet	LLURNET	OURS
1	89.72	86.22	88.34	89.62	86.65	85.93	**89.93**	86.44	89.48
2	80.92	95.06	80.27	**96.42**	84.62	88.56	81.22	87.10	86.49
3	37.39	56.03	35.49	59.03	65.55	61.34	72.77	**77.50**	56.95
4	65.43	63.18	57.40	57.33	70.30	**85.49**	82.54	84.82	84.04
5	61.99	99.20	63.30	**99.22**	85.40	87.95	84.81	85.02	77.37
6	50.35	60.96	57.35	66.02	74.82	**79.26**	75.11	69.09	75.92
7	77.38	83.18	74.24	83.43	81.53	64.75	78.74	82.31	**90.83**
AA	66.17	77.69	65.20	78.72	78.41	79.04	80.73	**81.75**	80.15
OA	72.41±3.04	80.42±3.40	70.50±5.71	80.96±4.56	79.81±2.78	80.27±1.38	81.94±1.55	83.53±1.17	**83.83±1.69**
kappa	66.52±3.69	76.32±4.18	64.15±6.96	76.92±5.59	75.47±3.11	76.81±1.92	78.33±2.47	80.17±1.26	**80.51±2.04**

TABLE V
CLASSIFICATION RESULTS (%) OF THE UP DATASET

Class	Deep coral	IRM	VREx	Mixstyle	SagNet	PDEN	SDEnet	LLURNET	OURS
1	83.42	85.92	86.21	**88.87**	86.68	69.06	87.4	87.08	71.44
2	73.24	67.23	78.26	58.78	**96.41**	87.72	95.55	90.63	91.62
3	43.83	39.86	41.99	45.46	33.38	**73.85**	25.91	36.94	51.71
4	55.11	77.05	48.33	**83.91**	19.02	28.12	9.02	33.46	23.38
5	79.50	92.64	84.33	64.10	99.68	84.69	99.47	**99.79**	87.65
6	61.36	73.00	63.03	**78.05**	52.93	57.13	61.04	60.93	67.54
7	57.80	44.19	61.14	34.70	**82.14**	80.73	75.60	73.35	68.74
AA	64.89	68.56	66.18	64.843	67.18	68.76	64.86	**68.88**	66.01
OA	63.21±4.90	66.86±4.31	65.21 ± 2.74	66.91±5.23	64.77±3.14	67.48±2.84	66.65±2.41	67.32±1.98	**69.56±2.17**
kappa	51.65±4.62	55.39±4.86	53.99 ± 2.69	55.80±6.54	52.74±3.98	57.51±3.26	55.98±2.79	56.97±2.32	**59.01±2.55**

TABLE VI
CLASSIFICATION RESULTS (%) OF THE HOUSTON DATASET

Class	Deep coral	IRM	VREx	Mixstyle	SagNet	PDEN	SDEnet	LLURNET	OURS
1	37.94	25.68	27.28	46.71	25.79	46.49	24.69	**78.79**	52.62
2	39.89	47.71	48.50	79.14	62.79	77.60	84.98	**87.48**	77.82
3	41.55	32.89	36.57	57.18	48.66	59.73	59.65	58.79	**63.88**
4	68.18	55.45	56.82	89.55	81.82	**100**	**100**	**100**	**100**
5	**87.29**	83.42	81.10	86.18	59.57	49.62	62.33	76.36	73.27
6	60.94	61.51	61.29	61.66	89.28	84.98	90.54	86.20	**92.33**
7	52.53	53.14	52.01	59.66	34.99	**64.21**	57.45	62.48	46.54
AA	55.47	51.40	51.94	68.58	57.56	68.95	68.52	**78.59**	72.35
OA	59.06±0.53	59.04±6.10	58.84±6.71	64.89±3.34	73.01±1.81	75.40±1.76	79.96±1.18	80.88±1.88	**81.12±3.13**
kappa	55.47±4.08	41.12±5.67	40.89±6.73	49.29±4.63	55.29±3.50	55.87±2.92	65.15±2.24	68.43±2.46	**72.35±3.52**

C. Experimental Results

From Tables IV to VII, we compare different methods with our proposed method on three public HSI datasets. By comparing the domain adaptation method Deep coral [47] with our DG method, we can observe that although domain adaptation methods have access to more prior information, they may not be suitable for hyperspectral scenes when applied to RGB images. In addition, these methods focus more on achieving distribution alignment between the source and target domains, which often overlooks the utilization of supervised information from the source domain and data from the target domain.

First, the method is compared with DG methods for RGB scenes, including IRM [24], VREx [43], Mixstyle [44],

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:59:23 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
CLASSIFICATION RESULTS (%) OF THE HYRANK DATASET

Class	Deep coral	IRM	VREx	Mixstyle	SagNet	PDEN	SDEnet	LLURNET	OURS
1	24.13	29.45	32.48	**44.13**	21.86	41.75	35.44	29.13	32.52
2	49.63	**95.06**	60.93	71.48	7.41	0.02	12.96	3.70	18.52
3	28.10	27.78	19.86	27.00	50.70	57.98	**62.68**	59.86	50.70
4	**64.05**	44.30	54.94	48.48	16.46	5.06	8.86	8.86	16.45
5	17.08	**30.32**	17.27	18.93	0.18	0.27	0.36	0.18	0.36
6	10.69	10.03	14.88	14.57	**33.65**	30.57	22.51	23.93	**33.65**
7	55.76	57.38	57.20	51.08	**81.81**	72.30	72.26	75.27	**81.81**
8	20.94	19.13	19.45	20.03	61.75	70.73	74.97	**77.34**	61.75
9	18.15	29.57	24.84	26.52	58.90	23.31	19.55	19.80	**71.43**
10	50.79	54.38	59.21	**61.02**	0.10	29.80	17.88	9.93	9.93
11	93.83	92.65	93.45	92.00	**100**	**100**	**100**	**100**	**100**
12	99.03	98.42	98.10	99.24	**100**	**100**	**100**	**100**	**100**
AA	44.35	49.04	46.05	47.87	44.39	44.32	43.96	42.33	**48.09**
OA	45.29±2.64	47.33±0.65	45.91 ± 1.67	44.96±2.53	61.87±1.67	61.52±2.01	61.65±1.23	62.48±1.58	**63.05±2.59**
kappa	36.75±2.24	38.82±0.82	37.56 ± 1.54	36.79±2.49	52.33±1.97	52.15±2.79	52.02±1.95	52.81±2.11	**54.22±3.87**

TABLE VIII
TRAINING TIME OF ONE EPOCH AND THE TESTING TIME FOR FOUR DATASETS

	Deep coral	IRM	VREx	Mixstyle	SagNet	PDEN	SDEnet	LLURNET	OURS
Dataset	PC(Target)								
Traing Time of one epoch	14.49	15.65	15.14	13.18	25.79	11.11	24.46	18.28	16.37
Testing Time	9.29	9.28	9.38	9.70	20.16	11.09	11.05	2.74	10.80
Dataset	UP(Target)								
Traing Time of one epoch	14.53	18.21	12.92	12.85	52.55	24.65	24.21	19.11	15.91
Testing Time	9.30	9.51	9.13	9.23	25.69	11.24	11.02	2.75	10.54
Dataset	HOUSTON2018(Target)								
Traing Time of one epoch	14.34	15.4	15.76	15.6	22.18	7.88	5.61	1.17	4.23
Testing Time	17.86	17.67	16.23	16.98	15.34	8.51	7.77	2.29	6.11
Dataset	LOUKIA(Target)								
Traing Time of one epoch	16.87	16.45	14.39	13.61	30.89	17.64	17.85	11.24	13.37
Testing Time	2.99	2.75	2.68	2.76	8.07	4.52	4.79	1.07	4.66

TABLE IX
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS

Val_{intra}	Val_{inter}	SG	HOUSTON18	PC	UP	Loukia
✓	✓		74.21±3.69	76.58±2.85	65.48±3.25	58.65±5.52
	✓	✓	79.82±3.57	80.21±2.80	68.11±2.97	62.57±4.23
✓		✓	80.33±2.45	82.98±1.47	68.32±2.72	62.96±3.01
✓	✓	✓	**81.12±3.13**	**83.83±1.69**	**69.56±2.17**	**63.05±2.59**

SagNet [45], and PDEN [46]. IRM divides the source domain into multiple domains. Since domain shift arises from within a single-source domain, its generalization performance on unknown target domains might be compromised. VREx aims to recover the causal mechanisms of the target domain. However, the complex causal relationships present in the target domain can limit the performance of the model. To mitigate style bias, MixStyle blends instance-level feature statistics from different source domains, while SagNet separates style encoding from category information. PDEN simulates photometric and geometric transformations in unknown target domains through domain expansion subnetworks. These methods primarily focus on RGB image classification and emphasize spatial information. In the case of HSIs, which contain rich spectral information compared with RGB images, these methods may not be applicable for single-source DG.

Then, the method is compared with DG methods for hyperspectral scenes, including SDEnet [8] and LLURNET [10]. SDEnet constructs a semantic encoder and a morphological encoder, where spatial randomization and spectral randomization are performed simultaneously in the semantic encoder

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:59:23 UTC from IEEE Xplore. Restrictions apply.

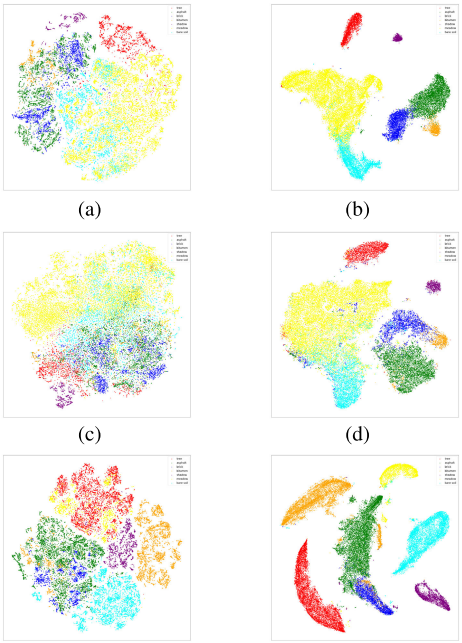


Fig. 10. t-SNE of class separability of the Pavia dataset, where SD is UP and TD is PC. (a) Raw sample of SD. (b) Features extracted from SD. (c) Generated samples. (d) Features extracted from generated samples. (e) Raw samples of TD. (f) Features extracted from TD.

to generate samples. However, SDEnet overly compresses the band information of HSIs. LLURnet designs a style randomization layer to achieve feature transformations and fusion. It uses an intraclass penalty term to avoid generating redundant samples. However, it overlooks the impact of feature transformations on category information. Our method leverages spectral unmixing to separate domain-invariant features from variable features. By randomizing the parameters of the variable features, we enhance the generalization performance of the model on unknown target domains. In addition, we introduce contrastive learning to ensure the effectiveness of generated samples. In comparison to the aforementioned methods, our proposed method demonstrates the highest accuracy. The classification prediction graphs of different methods are shown in Figs. 6–9; it can be found that our method is effective in improving the generalization performance on unknown domains.

D. Generalization Performance

The left side of Fig. 10 shows the distribution of the original samples in the source domain, extended domain, and target domain, while the right side shows the visualization results of the features extracted by the classifier. From Fig. 2, we can find that the generated domain not only differs from the distribution of the source domain but also retains class information in its low-dimensional representation. For unknown target domains, the features extracted by the classifier have separability, indicating that our method has a certain effect on improving the generalization performance of the model.

E. Ablation Experiment

We compared the training and testing times of different methods on various datasets in Table VIII. All the comparison

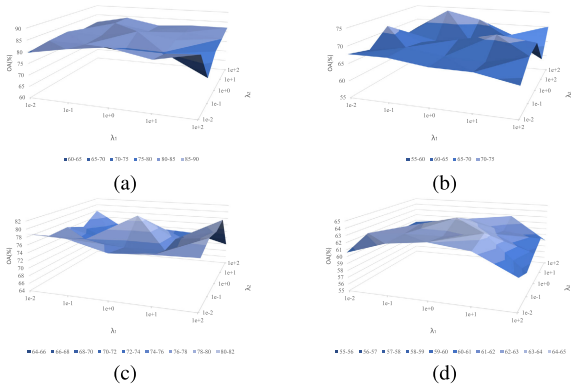


Fig. 11. Parameter tuning of λ_1 and λ_2 for four datasets. (a) PC. (b) UP. (c) Houston. (d) HyRANK.

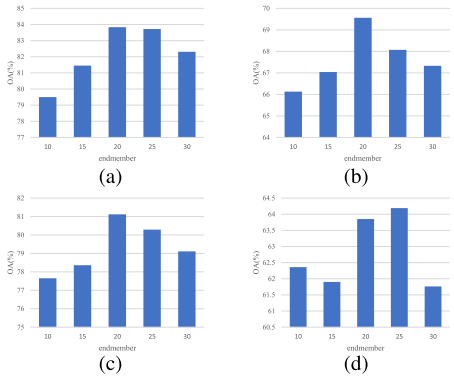


Fig. 12. Parameter tuning of endmember number for four datasets. (a) PC. (b) UP. (c) Houston. (d) HyRANK.

methods are trained using the same batch size 256 on NVIDIA GeForce RTX 2080 Ti. It was observed that the dataset size has a significant impact on the running time. In comparison to other methods, our approach achieved better results with a few extra time computation.

To evaluate the effectiveness of different modules, we conducted ablation studies using or removing these modules in Table IX. $\text{Val}_{\text{inter}}$ represents interdomain constraints, which are used to ensure differences between the generated samples and the source domain. $\text{Val}_{\text{intra}}$ represents intraclass constraints, which are used to ensure that the generated samples maintain their corresponding class during the process of generation. SG represents the sample generation. From Table IX, it can be seen that to a certain extent, introducing interdomain and interclass contrastive learning losses can ensure the effectiveness of generated samples, by maintaining differences from the source domain samples while having relatedness. The parameter randomization aims to simulate the phenomenon observed in the real world, where samples of the same class exhibit similar abundance matrices while having differences in their endmember information. It effectively improves the generalization performance of the model.

To evaluate the impact of different hyperparameter values on the accuracy of the target domain, we chose endmember number and the two hyperparameters λ_1 and λ_2 from (28) as adjustable parameters, with values $\{10, 15, 20, 25, 30\}$ and $\{1e-2, 1e-1, 1e+0, 1e+1, 1e+2\}$, respectively. Fig. 11 shows that when the hyperparameters λ_1 and λ_2 are too high or too low, the model's accuracy tends to decrease. As shown

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:59:23 UTC from IEEE Xplore. Restrictions apply.

in Fig. 12, when the endmember number is too large or too small, the model's accuracy tends to decrease. When the endmember number is too high, the model's sensitivity to noise increases, leading to unstable spectral unmixing results. On the other hand, when the endmember number is too low, the true components of mixed pixels may not be fully covered, resulting in information loss and inability to reflect the actual scene.

V. Conclusion

In this article, we propose a cross-domain hyperspectral classification network for single-source domain generation. To separate domain-invariant features from domain-specific features, we introduce a DG network based on nonlinear sample generation. We introduce a random environmental noise to simulate the differences between different domains and generate new samples. To ensure the effectiveness of the generated samples, contrastive learning is introduced. Interdomain contrastive learning is used to ensure differences between the generated domain and the source domain, while intradomain contrastive learning ensures relatedness. Several experiments have been conducted to demonstrate the effectiveness of our method.

References

[1] L. Liu, W. Li, Z. Shi, and Z. Zou, "Physics-informed hyperspectral remote sensing image synthesis with deep conditional generative adversarial networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5528215.

[2] R. Hang, X. Qian, and Q. Liu, "Cross-modality contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5532812.

[3] C.-I. Chang, "Hyperspectral anomaly detection: A dual theory of hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511720.

[4] J. Kang et al., "DisOptNet: Distilling semantic knowledge from optical images for weather-independent building segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4706315.

[5] Y. Su et al., "ACGT-Net: Adaptive cuckoo refinement-based graph transfer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5521314.

[6] R. Hang, P. Yang, F. Zhou, and Q. Liu, "Multiscale progressive segmentation network for high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5412012.

[7] J. Wang et al., "Generalizing to unseen domains: A survey on domain generalization," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 8, pp. 8052–8072, Aug. 2023.

[8] Y. Zhang, W. Li, W. Sun, R. Tao, and Q. Du, "Single-source domain expansion network for cross-scene hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 32, pp. 1498–1512, 2023.

[9] Y. Zhang, M. Zhang, W. Li, S. Wang, and R. Tao, "Language-aware domain generalization network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501312.

[10] H. Zhao, J. Zhang, L. Lin, J. Wang, S. Gao, and Z. Zhang, "Locally linear unbiased randomization network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501312.

[11] M. Zhao, M. Wang, J. Chen, and S. Rahardja, "Hyperspectral unmixing for additive nonlinear models with a 3-D-CNN autoencoder network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5509415.

[12] S. Mei, R. Jiang, X. Li, and Q. Du, "Spatial and spectral joint super-resolution using convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4590–4603, Jul. 2020.

[13] S. Mei, Y. Geng, J. Hou, and Q. Du, "Learning hyperspectral images from RGB images via a coarse-to-fine CNN," *Sci. China Inf. Sci.*, vol. 65, no. 5, pp. 1–14, May 2022.

[14] S. Mei et al., "Lightweight multiresolution feature fusion network for spectral super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501414.

[15] R. Hang, Q. Liu, and Z. Li, "Spectral super-resolution network guided by intrinsic properties of hyperspectral imagery," *IEEE Trans. Image Process.*, vol. 30, pp. 7256–7265, 2021.

[16] X. Peng, F. Qiao, and L. Zhao, "Out-of-domain generalization from a single source: An uncertainty quantification approach," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 3, pp. 1775–1787, 2022.

[17] K. Zhou, Y. Yang, T. Hospedales, and T. Xiang, "Deep domain-adversarial image generation for domain generalisation," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 7, pp. 13025–13032.

[18] N. Somavarapu, C.-Y. Ma, and Z. Kira, "Frustratingly simple domain generalization via image stylization," 2020, *arXiv:2006.11207*.

[19] Y. N. D. H. Zhang, M. Cisse, and D. Lopez-Paz, "mixup: Beyond empirical risk minimization," in *Proc. Int. Conf. Learn. Represent.*, 2018, pp. 1–9.

[20] H. Yu et al., "Global spatial and local spectral similarity-based manifold learning group sparse representation for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3043–3056, May 2020.

[21] Y. Su, L. Gao, M. Jiang, A. Plaza, X. Sun, and B. Zhang, "NSCKL: Normalized spectral clustering with kernel-based learning for semisupervised hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 53, no. 10, pp. 6649–6662, Oct. 2023.

[22] Y. Li, M. Gong, X. Tian, T. Liu, and D. Tao, "Domain generalization via conditional invariant representations," in *Proc. AAAI Conf. Artif. Intell.*, 2018, vol. 32, no. 1, pp. 3579–3587.

[23] M. M. Rahman, C. Fookes, M. Baktashmotlagh, and S. Sridharan, "Correlation-aware adversarial domain adaptation and generalization," *Pattern Recognit.*, vol. 100, Apr. 2020, Art. no. 107124.

[24] M. Arjovsky, L. Bottou, I. Gulrajani, and D. Lopez-Paz, "Invariant risk minimization," 2019, *arXiv:1907.02893*.

[25] V. Piratla, P. Netrapalli, and S. Sarawagi, "Efficient domain generalization via common-specific low-rank decomposition," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 7728–7738.

[26] H. Zhang, Y.-F. Zhang, W. Liu, A. Weller, B. Scholkopf, and E. P. Xing, "Towards principled disentanglement for domain generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 8024–8034.

[27] C. Liu et al., "Learning causal semantic representation for out-of-distribution prediction," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 34, 2021, pp. 6155–6170.

[28] K. Zhou, Y. Yang, Y. Qiao, and T. Xiang, "Domain adaptive ensemble learning," *IEEE Trans. Image Process.*, vol. 30, pp. 8008–8018, 2021.

[29] D. Kim, Y. Yoo, S. Park, J. Kim, and J. Lee, "SelfReg: Self-supervised contrastive regularization for domain generalization," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9619–9628.

[30] M. Narayanan, V. Rajendran, and B. Kimia, "Shape-biased domain generalization via shock graph embeddings," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 1315–1325.

[31] Y. Qian, S. Jia, J. Zhou, and A. Robles-Kelly, "Hyperspectral unmixing via $L_{1/2}$ sparsity-constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4282–4297, Nov. 2011.

[32] S. Zhang et al., "Spectral–spatial hyperspectral unmixing using nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5505713.

[33] D. Hong et al., "Endmember-guided unmixing network (EGU-Net): A general deep learning framework for self-supervised hyperspectral unmixing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6518–6531, Nov. 2022.

[34] J. V. Stone, "Independent component analysis: An introduction," *Trends Cogn. Sci.*, vol. 6, no. 2, pp. 59–64, 2002.

[35] J. Yao, D. Meng, Q. Zhao, W. Cao, and Z. Xu, "Nonconvex-sparsity and nonlocal-smoothness-based blind hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 6, pp. 2991–3006, Jun. 2019.

[36] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2018.

[37] C. C. Borel and S. A. W. Gerstl, "Nonlinear spectral mixing models for vegetative and soil surfaces," *Remote Sens. Environ.*, vol. 47, no. 3, pp. 403–416, Mar. 1994.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:59:23 UTC from IEEE Xplore. Restrictions apply.

[38] G. A. Licciardi and F. Del Frate, "Pixel unmixing in hyperspectral data by means of neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4163–4172, Nov. 2011.
[39] B. Palsson, J. Sigurdsson, J. R. Sveinsson, and M. O. Ulfarsson, "Hyperspectral unmixing using a neural network autoencoder," *IEEE Access*, vol. 6, pp. 25646–25656, 2018.
[40] L. Qi, F. Gao, J. Dong, X. Gao, and Q. Du, "SSCU-Net: Spatial–spectral collaborative unmixing network for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5407515.
[41] B. Palsson, M. O. Ulfarsson, and J. R. Sveinsson, "Convolutional autoencoder for spectral–spatial hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 535–549, Jan. 2020.
[42] R. K. Baghbaderani, Y. Qu, and H. Qi, "Unsupervised hyperspectral image domain adaptation through unmixing-based domain alignment," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2023, pp. 5906–5909.
[43] D. Krueger et al., "Out-of-distribution generalization via risk extrapolation (REx)," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 5815–5826.
[44] K. Zhou, Y. Yang, Y. Qiao, and T. Xiang, "Domain generalization with MixStyle," 2021, *arXiv:2104.02008*.
[45] H. Nam, H. Lee, J. Park, W. Yoon, and D. Yoo, "Reducing domain gap by reducing style bias," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8690–8699.
[46] L. Li et al., "Progressive domain expansion network for single domain generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 224–233.
[47] B. Sun and K. Saenko, "Deep CORAL: Correlation alignment for deep domain adaptation," in *Proc. Eur. Conf. Comput. Vis. ECCV Workshops*. Amsterdam, The Netherlands: Springer, Oct. 2016, pp. 443–450.



Biqi Wang (Graduate Student Member, IEEE) received the B.Admin. degree in information management and information system from the School of Information Engineering, Nanjing University of Finance and Economics, Nanjing, China, in 2019. She is currently pursuing the Ph.D. degree with the Nanjing University of Science and Technology, Nanjing.

Her research interests are transfer learning and hyperspectral image processing.



Yang Xu (Member, IEEE) received the B.Sc. degree in applied mathematics and the Ph.D. degree in pattern recognition and intelligence systems from Nanjing University of Science and Technology (NUST), Nanjing, China, in 2011 and 2016, respectively.

He is currently an Associate Professor with the School of Computer Science and Engineering, NUST. His research interests include hyperspectral image classification, hyperspectral detection, image processing, and machine learning.



Zebin Wu (Senior Member, IEEE) received the B.Sc. and Ph.D. degrees in computer science and technology from the Nanjing University of Science and Technology, Nanjing, China, in 2003 and 2007, respectively.

He is currently a Professor with the School of Computer Science and Engineering, Nanjing University of Science and Technology. He was a Visiting Scholar with the GIPSA-lab, Grenoble INP, the University Grenoble Alpes, Grenoble, France, from August 2018 to September 2018. He was a Visiting Scholar at the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, Escuela Politécnica, University of Extremadura, Cáceres, Spain, from June 2014 to June 2015. He was a Visiting Scholar at the Department of Mathematics, University of California Los Angeles, Los Angeles, CA, USA, from August 2016 to September 2016 and from July 2017 to August 2017. His research interests include hyperspectral image processing, parallel computing, and big data processing.



Shangdong Zheng was born in Jiangsu, China, in 1995. He received the B.S. degree in software engineering from the School of Computer and Communication, Lanzhou University of Technology (LUT), Lanzhou, Gansu, China, in 2017, and the Ph.D. degree in computer science and technology from the Nanjing University of Science and Technology (NJUST), Nanjing, Jiangsu, China, in 2023.

He is currently a Lecturer with the Nanjing University of Posts and Telecommunications (NJUPT), Nanjing, China. His research interests include image processing, object detection, and machine learning.



Zhihui Wei (Member, IEEE) was born in Jiangsu, China, in 1963. He received the B.Sc. and M.Sc. degrees in applied mathematics and the Ph.D. degree in communication and information system from South East University, Nanjing, China, in 1983, 1986, and 2003, respectively.

He is currently a Professor and a Ph.D. Supervisor with the Nanjing University of Science and Technology (NUST), Nanjing. His research interests include partial differential equations, mathematical image processing, multiscale analysis, sparse representation, and compressive sensing.



Jocelyn Chanussot (Fellow, IEEE) received the M.Sc. degree in electrical engineering from the Grenoble Institute of Technology (Grenoble INP), Grenoble, France, in 1995, and the Ph.D. degree from the Université de Savoie, Annecy, France, in 1998.

Since 1999, he has been with Grenoble INP. He was a Visiting Scholar with Stanford University, Stanford, CA, USA, KTH, Stockholm, Sweden, and the National University of Singapore (NUS),Singapore. Since 2013, he has been an Adjunct Professor with the University of Iceland, Reykjavik, Iceland. From 2015 to 2017,he was a Visiting Professor with the University of California at Los Angeles(UCLA), Los Angeles, CA, USA. He holds the AXA Chair in remote sensing and is an Adjunct Professor with the Chinese Academy of Sciences, Aerospace Information Research Institute, Beijing, China. He is a Professor of signal and image processing with Grenoble INP. His research interests include image analysis, hyperspectral remote sensing, data fusion, machine learning, and artificial intelligence.

Dr. Chanussot was a member of the Machine Learning for Signal Processing Technical Committee of the IEEE Signal Processing Society from 2006 to 2008, and the Institut Universitaire de France from 2012 to 2017.He was the Founding President of the IEEE Geoscience and Remote Sensing (GRS) Society French Chapter from 2007 to 2010, which received the 2010 IEEE GRSS Chapter Excellence Award. He is a Highly Cited Researcher (Clarivate Analytics, Thomson Reuters) from 2018 to 2019. He was the Vice-President of the IEEE GRS Society (GRSS) and the in-charge of meetings and symposia, from 2017 to 2019. He was the General Chair of the first IEEE GRSS Workshop on Hyperspectral Image and Signal Processing, Evolution in Remote Sensing (WHISPERS). He was the Chair from 2009 to2011 and the Co-Chair of the GRSS Data Fusion Technical Committee from 2005 to 2008. He was the Program Chair of the IEEE International Workshop on Machine Learning for Signal Processing in 2009. He was the Editor-in-Chief of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING from 2011 to 2015. He is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and IEEE TRANSACTIONS ON IMAGE PROCESSING, AND PROCEEDINGS.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:59:23 UTC from IEEE Xplore. Restrictions apply.